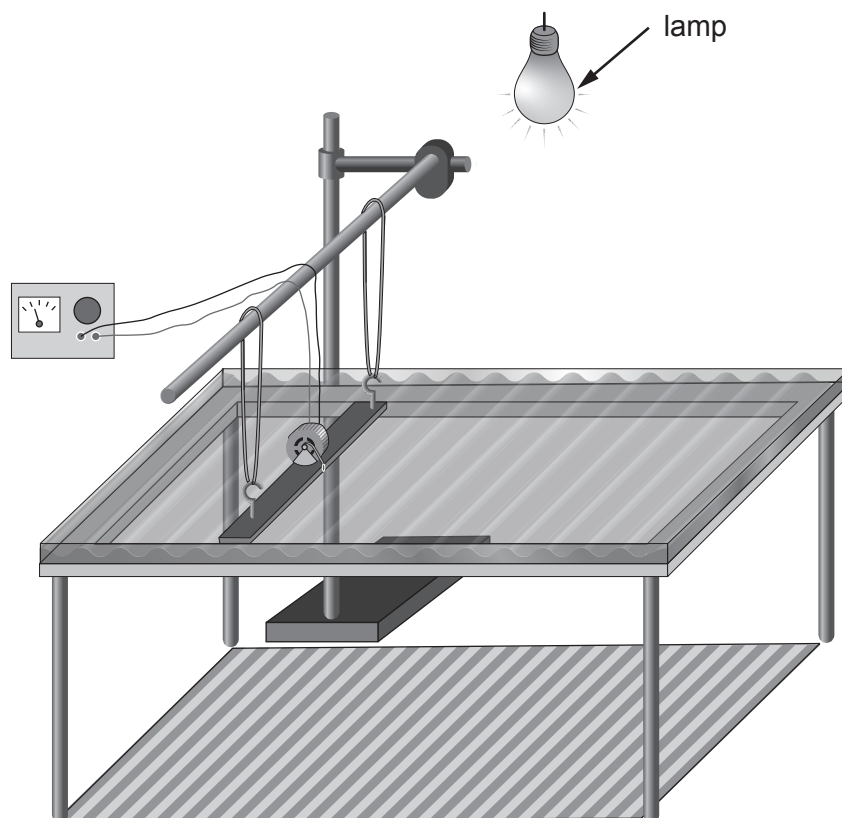


2. A teacher demonstrates waves using a ripple tank. She changes the frequency of the waves produced and the class observes the effect on their wavelength.



Next, the class investigates the link between frequency and wavelength using a virtual ripple tank simulation.

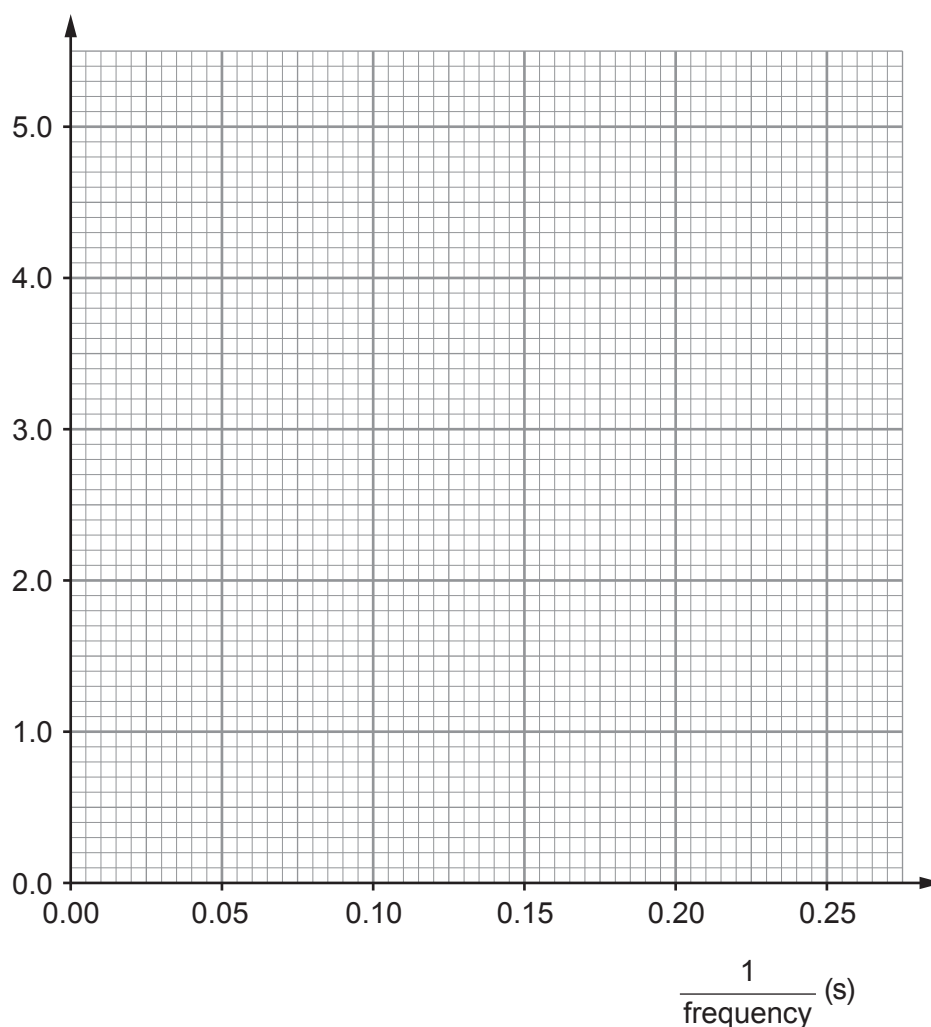
The table below shows their results.

| Frequency (Hz) | $\frac{1}{\text{frequency}}$ (s) | Wavelength (cm) |
|----------------|----------------------------------|-----------------|
| 20.0           | 0.05                             | 1.0             |
| 10.0           | 0.10                             | 2.0             |
| 6.7            | 0.15                             | 3.0             |
| 5.0            | 0.20                             | 4.0             |
| 4.0            | 0.25                             | 5.0             |



- (a) (i) Plot the data on the grid below and draw a suitable straight line. [3]

Wavelength (cm)



- (ii) **Complete** the following sentences by underlining the correct word or phrase in the brackets. [2]

- I. As  $\frac{1}{\text{frequency}}$  increases the wavelength [**increases / decreases / stays the same**].
- II. As frequency increases the wavelength [**increases / decreases / stays the same**].



- (iii) I. Use data from the table to state the wavelength of the waves at a frequency of **10 Hz**. [1]

Wavelength = ..... cm

- II. Use the equation:

$$\text{wave speed} = \text{frequency} \times \text{wavelength}$$

to determine the speed of the waves at a frequency of **10 Hz**. [2]

Speed = ..... cm/s

- (b) (i) Electromagnetic waves are used to communicate with satellites. Some satellites remain above the same point on the Earth to allow constant communication.

**Complete** the following sentences about communications satellites by underlining the correct word or phrase in the brackets. [4]

Electromagnetic waves are (**longitudinal / parallel / transverse**) waves.

TV signals are sent to satellites in (**geothermal / geosynchronous / geostationary**) orbits using (**microwaves / visible light / gamma rays**).

These satellites orbit above the (**poles / equator / axis**) of the Earth.

- (ii) A satellite orbits the Earth in a circular orbit, once every 24 hours. The radius of its orbit is 42 164 km.

- I. Use the equation:

$$\text{circumference of a circle} = 2\pi r \quad (\text{where } r = \text{radius and } \pi = 3.14)$$

to calculate the distance the satellite travels in one orbit. [1]

Distance = ..... km



II. Use the equation:

$$\text{speed} = \frac{\text{distance}}{\text{time}}$$

to calculate the speed of the satellite in km/h.

[2]

Speed = ..... km/h

- (iii) Maddie suggests that the satellite orbits at the same speed as a point on the Earth's surface moves, so that it always stays above the same point on the Earth. Explain whether or not you agree. [2]

.....

.....

.....

3430U301  
07

17

